

Idunnuoluwa A. Adeniji

Chicago, IL | idunnuoluwaadeniji@gmail.com | [LinkedIn](#) | [Website](#)

Summary

Ph.D. student in Computer Science at UIC with a B.Sc. in Computational Science and Engineering. I am skilled in scientific visualization, VR/AR/MR, and interactive tool development. Research focuses on building accessible visualization tools for science and engineering.

Technical Skills

Languages & Tools: Python, Java, C++, C#, R, JavaScript, HTML, Unity, Babylon.js, ParaView, Maya, Adobe Photoshop
Specializations: Scientific visualization, VR/AR/MR, WebXR, 3D rendering, Data analysis, Human–Computer Interaction
Core Strengths: Critical thinking, Problem solving, Team collaboration, Adequate Communication, Time management

Professional Experience

Research Assistant – University of Illinois Chicago, Electronic Visualization Laboratory | 2025 – Present

- Responsible for system stability, performance, and ongoing development of the **OpenCosmo Visualization application**.
- Leads architectural decisions and maintenance of the rendering pipeline to support large-scale interactive data visualization in the browser.
- Conducts performance-focused research on WebGPU-based rendering and interface redesign.

Research Aide – ALCF Visualization Team, Argonne National Laboratory | 2025

- Designed and developed an **interactive GPU accelerated 3D web-based viewer** using Babylon.js for efficient scientific exploration of large-scale cosmological data
- Collaborated with the cosmology team to integrate domain-specific visualization features
- Presented results at the **Division poster session** and the **eScience conference**

Research Aide – ALCF Visualization Team, Argonne National Laboratory | 2023, 2024

- Created VR interactions for **halo datasets** and built immersive virtual environments for efficient data analysis
- Applied **ParaView** and **Unity** for visualization of high-performance cosmology simulations
- Presented research at the **Sandia XR Conference** and **Division poster session**

Student Assistant – Visualization and Immersive Studio for Education and Research (VISER), Kean University | 2022–2024

- Designed and managed visualization projects using **Zignage** and **Adobe Photoshop**
- Supported troubleshooting and daily operations of the VISER lab
- Assisted in public demonstrations and outreach activities

Teaching Assistant – University of Illinois at Chicago | 2025

- Led two lab sections for **CS 107 (Intro to Computing and Programming)**
- Guided students in debugging, problem-solving, and project development
- Held office hours, graded assignments, and proctored exams

Education

University of Illinois at Chicago | 2025 – Present

Ph.D. in Computer Science

Kean University | 2021 – 2024

B.Sc. in Computational Science and Engineering, **Summa Cum Laude**

Dean’s List (SP 2022 – FA 2024)

Projects & Publications

- Exploring Large-Scale Scientific Data in Virtual Reality** – Developed a Unity-based tool for HACC cosmology data exploration
- Interactive Exploration of HACC Data Using WebXR** – Developed a Web-based visualization tool to be integrated in the ALCF HACC Compute Portal. ([OpenCosmo Visualization App](#))
- Guardian of the Sea** – Visualization project inspired by the UN Sustainable Development Goal *Life Below Water* aimed to educate users on the dangers of pollution and ways to mitigate it.
- Publication:** *Using Unity for Scientific Visualization as a Course-based Undergraduate Research Experience* ([link](#))

Leadership & Activities

- Presenter: Sandia XR Conference, eScience, CAAV, KEAN Research Days, CELS Division Poster session
- Member: ACM Kean Chapter, Women in Cybersecurity, National Society of Collegiate Scholars
- Scholarships: Kean STEM Scholarship, Bodzioch Advancement of Science Scholarship
- Outreach: Argonne coding camp volunteer, Tutoring Chicago math tutor